433 MHz Wide-Sector Antenna

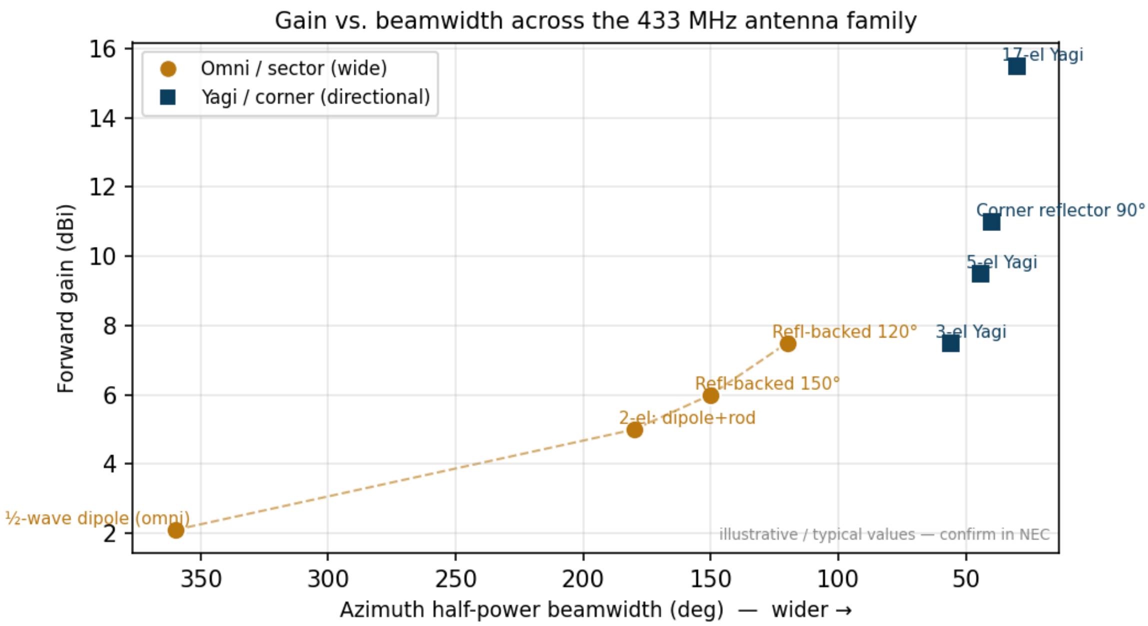
Reflector-backed dipole — detailed construction, family comparison & patterns
433.92 MHz | λ = 690.9 mm | detailed build = 120° version

Specification (120° build)	Value
Type	Folded ½-wave dipole + flat/grid reflector (sector panel)
Azimuth beamwidth	~120° (tunable 120–180°)
Forward gain / F-B	≈ 7.5 dBi / ~15 dB
Driven element	folded dipole 327 mm, 6 mm tube, 138 mm in front of reflector
Reflector	0.7λ×0.7λ = 484×484 mm plate or vertical-rod grid
Feed / polarisation	50 Ω + 1:1 choke balun / dipole vertical → horizontal sector

1 Antenna-family comparison

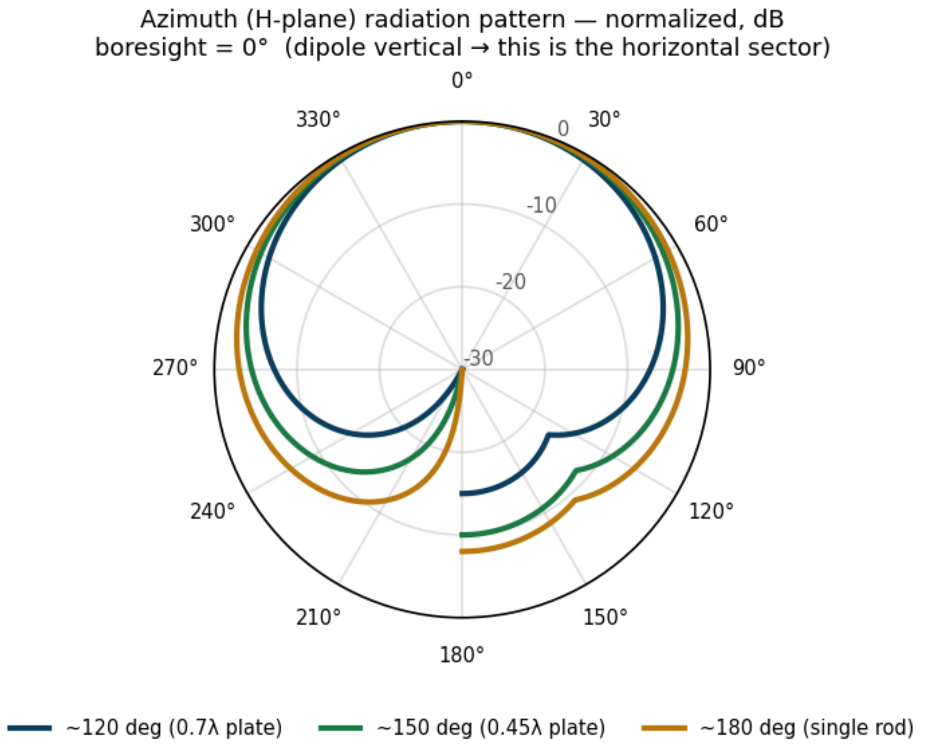
Where this design sits among 433 MHz options — from omni to high-gain pencil beam. Wider azimuth coverage costs gain; more gain narrows the beam in *both* planes.

Antenna	Az HPBW	El HPBW	Gain	F/B	Size / boom	Build
½-wave dipole (omni)	360°	~78°	~2.1 dBi	—	tiny	trivial
2-el: dipole + rod reflector	~180°	~75°	~5 dBi	~8 dB	small	easy
Reflector-backed dipole 150°	~150°	~70°	~6 dBi	~10 dB	medium	easy
Reflector-backed dipole 120°	**~120°**	~65°	**~7.5 dBi**	~15 dB	medium	easy
3-element Yagi	~56°	~63°	~7.5 dBi	~12 dB	0.24 m	easy
5-element Yagi	~44°	~52°	~9.5 dBi	~15 dB	0.8 m	moderate
Corner reflector (90°)	~40°	~45°	~11 dBi	~20 dB	medium	moderate
17-element Yagi	~30°	~34°	~15.5 dBi	~20 dB	3.0 m	hard

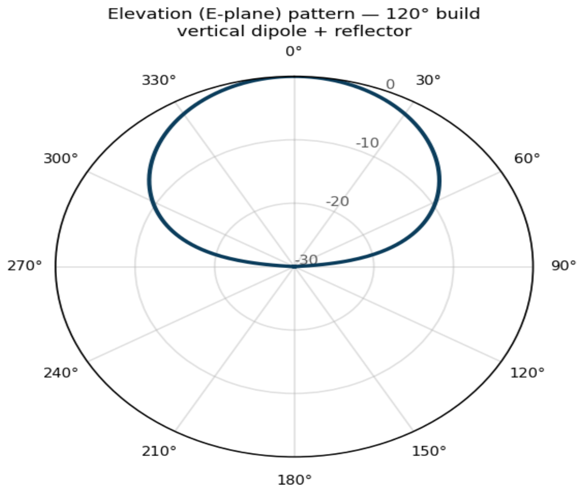


Gain vs. azimuth beamwidth. Orange = omni/sector family (this design's class); navy = directional Yagi/corner. The sector family trades beamwidth for gain along the dashed line.

2 Radiation patterns



Azimuth (horizontal) cut for the three reflector options — wider reflector narrows the sector and deepens the rear null; a single rod gives the ~180° cardioid.



Elevation (vertical) cut, 120° build — a vertical dipole holds the beam near the horizon while the reflector suppresses the rear.

3 Detailed construction

3.1 Tools required

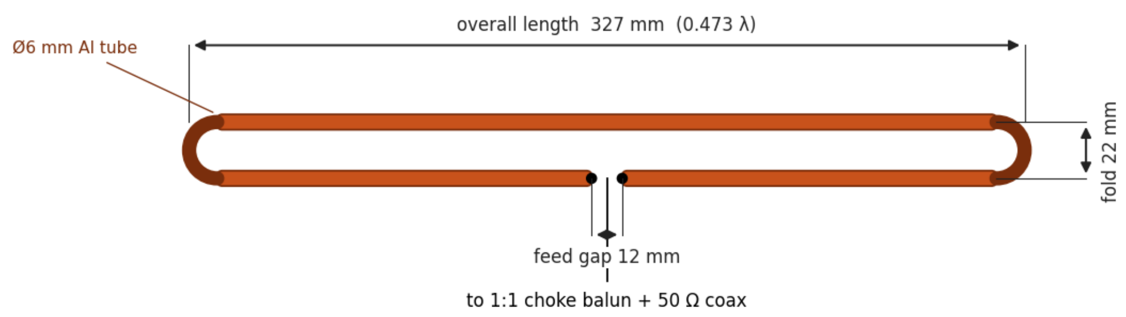
Cutting / forming	Measuring / tuning	Joining / finishing
hacksaw or tube cutter; tube bender or vice + former; file/deburr tool; drill + bits	steel tape / digital caliper; marker; **VNA or SWR/antenna analyser** (essential for tuning); multimeter (continuity)	soldering iron (60–80 W) or crimp tool; screwdrivers; spanners; heat gun

3.2 Bill of materials (detailed)

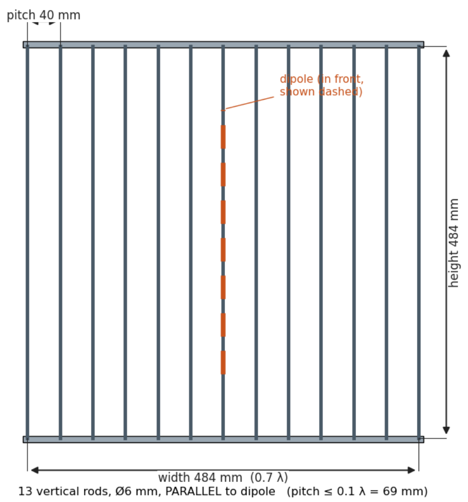
#	Item	Spec / example	Qty
1	Dipole tube	6 mm aluminium tube, ~377 mm stock to form a 327 mm folded loop	1
2	Reflector plate	aluminium sheet 1–2 mm, 484×484 mm — OR → item 3	1
3	Reflector grid (alt.)	~13× 6 mm Al rod, 484 mm long, on two cross-spreaders	1 set
4	Standoffs	nylon/PVC, 138 mm long, M5/M6	2–4
5	Support boom/arm	PVC 25 mm or fibreglass, dipole-to-mast	1
6	Coax connector	SMA or N female, bulkhead/flange, PTFE	1
7	Choke balun	RG-316/RG-58 coil OR type-61 ferrites OR λ/4 sleeve	1
8	Feed line	50 Ω coax: RG-316/RG-58 (short) or LMR-240/RG-8X (run)	as needed
9	Hardware	stainless M3/M5 screws, nylocks, solder lugs, ring tags	—
10	Weatherproofing	self-amalgamating tape, RTV silicone, heat-shrink, UV cable ties	—

3.3 Mechanical drawings

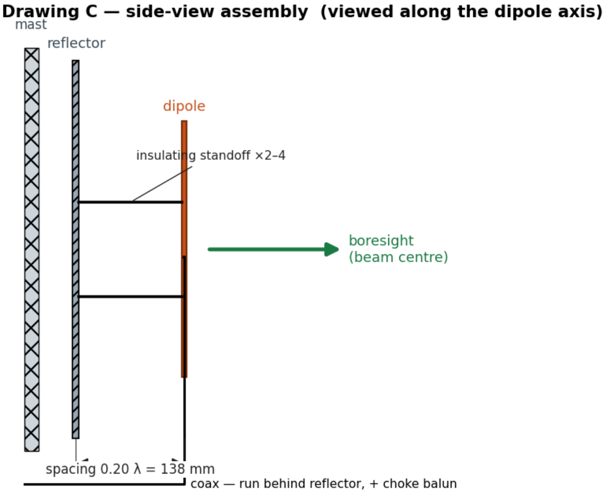
Drawing A — folded ½-wave dipole (front view)



Drawing B — reflector, wire-grid (front view)

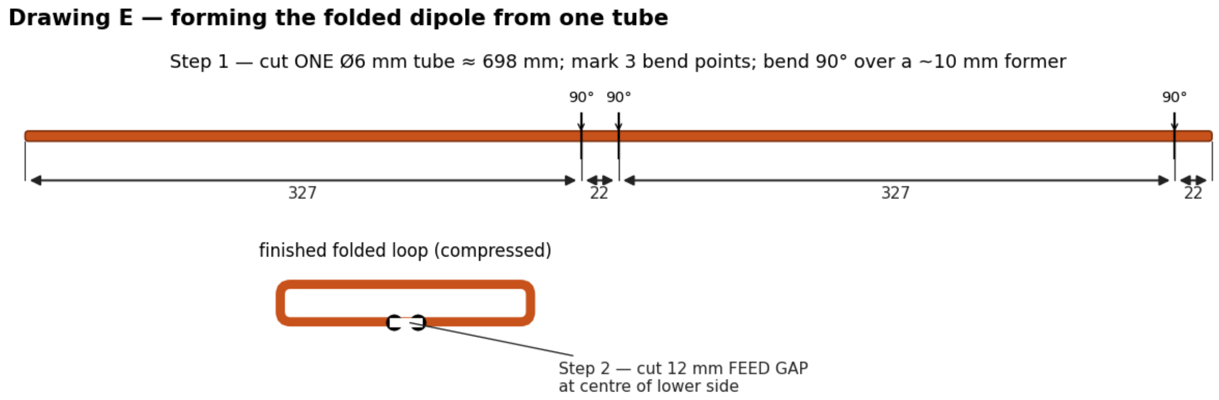


Drawing C — side-view assembly (viewed along the dipole axis)

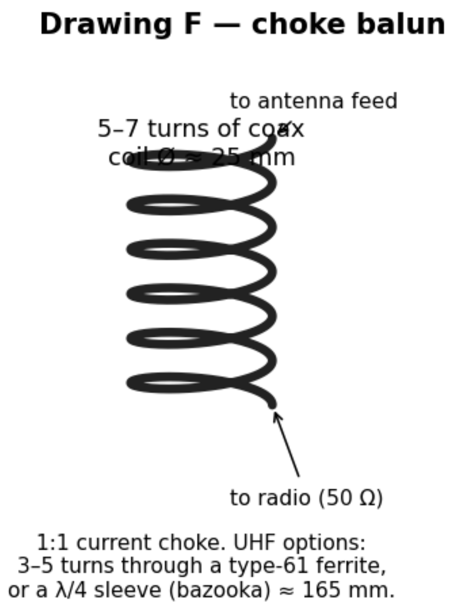


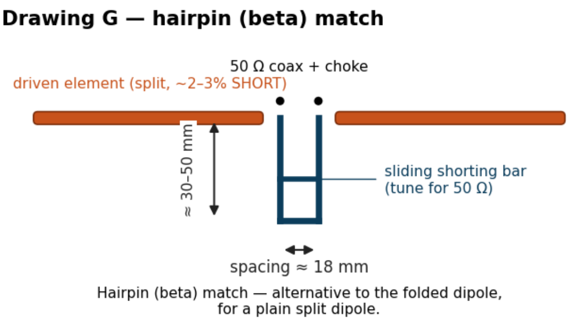
Drawings A–C: folded-dipole detail, reflector-grid front view, side-view assembly.

3.3 Mechanical drawings (continued)



Drawing E — forming the folded dipole from a single tube (Section 3.4).





Drawings F & G — choke balun, and the hairpin-match alternative (Section 3.7).

3.4 Building the folded dipole

1. Cut ~377 mm of 6 mm tube. Mark the four bend points per Drawing E: 327 mm → 22 mm → 327 mm → 22 mm.
2. Bend each mark 90° over a ~10–12 mm radius former (a socket or pipe) so the tube doesn’t kink; finish as a thin rectangular loop ~22 mm tall.
3. The two free ends meet at the centre of the lower long side — leave a **12 mm feed gap** between them; file the ends flat.
4. Fit a solder lug / ring tag to each end at the gap. The folded geometry must stay flat and symmetric (both halves equal).
5. Keep the loop in one plane; twist or skew here shows up as poor SWR and a tilted pattern.

3.5 Building the reflector

- **Solid plate:** cut aluminium sheet to 484×484 mm; deburr edges; drill centre holes for the standoffs.
- **Wire grid (lighter, ~same RF, less wind load):** ~13 vertical 6 mm rods, each 484 mm long, at 40 mm pitch (≤ 0.1λ = 69 mm). Rods **parallel to the dipole (vertical)**; bond them to two horizontal spreaders.
- Either way the reflector must be flat and electrically continuous (bonded joints). A bigger reflector narrows the beam and raises F/B; a smaller one widens it (Section 1 table).

3.6 Mounting & setting the spacing

- Fix the dipole 138 mm in front of the reflector **centre**, on 2–4 nylon/PVC standoffs (Drawing C). This 0.20λ spacing is the key set-up dimension.
- Orient the dipole **vertical** so the wide lobe is in azimuth (horizontal). Keep it parallel to the reflector plane and centred.

- Mount the support boom to the mast **behind** the reflector; keep all metalwork out of the forward field.
- Spacing is adjustable 104–173 mm (0.15–0.25λ): closer = wider beam & lower F/B; farther = narrower & higher F/B.

3.7 Feed, balun & matching

Folded dipole + 1:1 choke (recommended). Solder the coax centre to one feed-gap lug and the braid to the other. Immediately add a **1:1 current balun** (Drawing F): an air-core coax choke of 5–7 turns at ~25 mm diameter, OR 3–5 turns through a type-61 ferrite, OR a λ/4 sleeve (bazooka) balun ~164 mm. The choke blocks shield current that would otherwise radiate and broaden/skew the pattern.

Hairpin (beta) match — alternative. If you use a plain (un-folded) split dipole, its feed is low (~20–35 Ω). Make it ~2–3% short (capacitive) and bridge a shorted U-shaped **hairpin** across the gap (Drawing G): ~15 mm leg spacing, ~30–50 mm long of 4–6 mm rod; slide a shorting bar along it for 50 Ω. Still add a 1:1 choke.

3.8 Weatherproofing & grounding

- Seal the connector/feed: heat-shrink, then self-amalgamating tape, then a skin of RTV silicone. Leave a downward **drip loop** in the coax.
- If using an enclosure/radome, add a small **drain hole** at the lowest point; never trap condensation.
- Bond the reflector/mast to a ground/earth for lightning safety; for transmit, a coax surge arrestor at the building entry is good practice.
- Use stainless hardware and anti-oxidant on aluminium joints to stop galvanic corrosion outdoors.

3.9 Tuning procedure (with a VNA)

1. Mount the antenna in the clear (≥ 1–2 m from metal/ground) pointing at open space; connect the VNA at the coax end you'll actually use.
2. Sweep 400–470 MHz. Find the SWR minimum (the resonance dip).
3. If the dip is **below** 434 MHz → the dipole is too long: trim 2–3 mm per side and re-sweep. If **above** → it's too short: add length (or bend tips out).
4. Once the dip sits at ~434 MHz, fine-tune the **match**: nudge reflector spacing (folded dipole) or the hairpin shorting bar (hairpin) for SWR < 1.5:1.
5. Confirm the choke works: grip/move the coax — if SWR shifts, add more choke turns/ferrites.

3.10 Troubleshooting

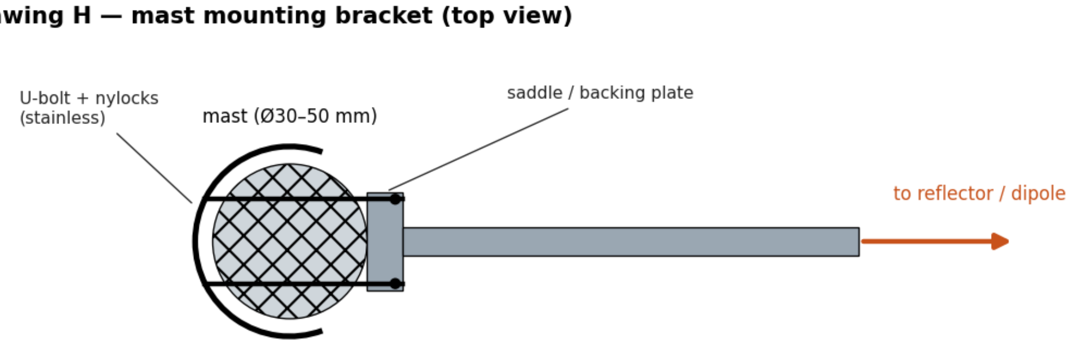
Symptom	Likely cause	Fix
SWR minimum too high in frequency	dipole too short	add length / bend tips outward
SWR minimum too low in frequency	dipole too long	trim 2–3 mm per side
SWR never below ~2:1	impedance mismatch / no choke	add/adjust hairpin or use folded dipole; fit choke
SWR changes when you touch the coax	common-mode current	more choke turns / add ferrites
Beam tilted or weak front-to-back	skewed dipole or coax in the field	re-square the dipole; route coax behind reflector
Pattern too narrow / too wide	reflector size/spacing off	resize reflector or change spacing (Sec.1)

3.11 Tolerances & gotchas

- Lengths ±1 mm, spacing ±3 mm. Spacing mostly trades beamwidth/F-B, not centre frequency.
- Grid rods **must be parallel to the dipole**; perpendicular rods barely reflect.
- Always choke the feed — an un-choked coax radiates and smears the sector.
- Keep mast/clamps behind the reflector plane; keep the dipole flat and centred.
- Aluminium doesn't solder easily — use stainless screws + solder lugs, or crimp ring tags, at joints.

4 Mounting & printable templates

4.1 Mast bracket

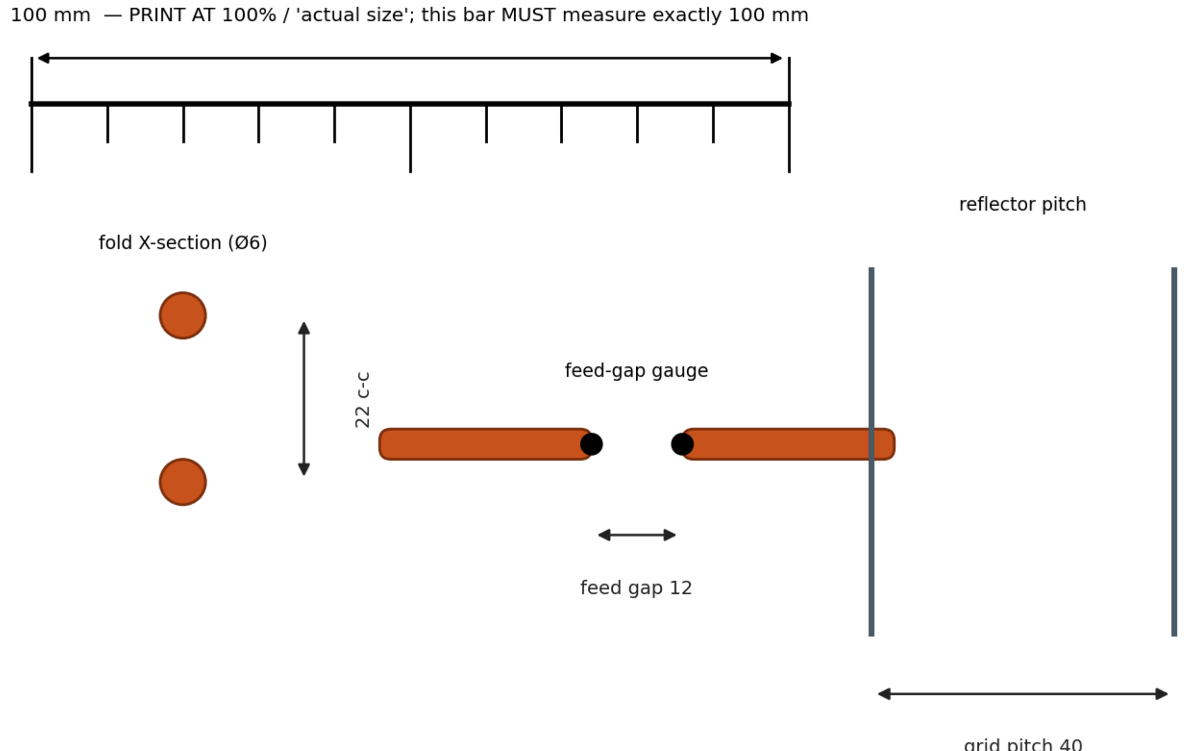


Drawing H — the support boom clamps to the mast with a stainless U-bolt and saddle/backing plate; the boom carries the reflector+dipole forward. Keep the mast behind the reflector plane.

- Use a **stainless U-bolt** sized to the mast (Ø30–50 mm) with a saddle and nylock nuts.
- Boom: 25 mm PVC or fibreglass; bolt the reflector to its forward end so the mast stays out of the beam.
- For down-tilt, shim the bracket a few degrees nose-down (useful when covering ground below the mast).

4.2 1:1 printable templates

Print this page (or just the gauges) at **100% / "actual size"** (no "fit to page"). First confirm the 100 mm calibration bar measures exactly 100 mm with a ruler — if not, adjust the printer scale until it does, then cut out the gauges.



Drawing J — 1:1 detail gauges (cut these out as go/no-go checks). Full 327 mm element: mark with a steel tape per the table.

4.3 Marking table (mark with a steel tape)

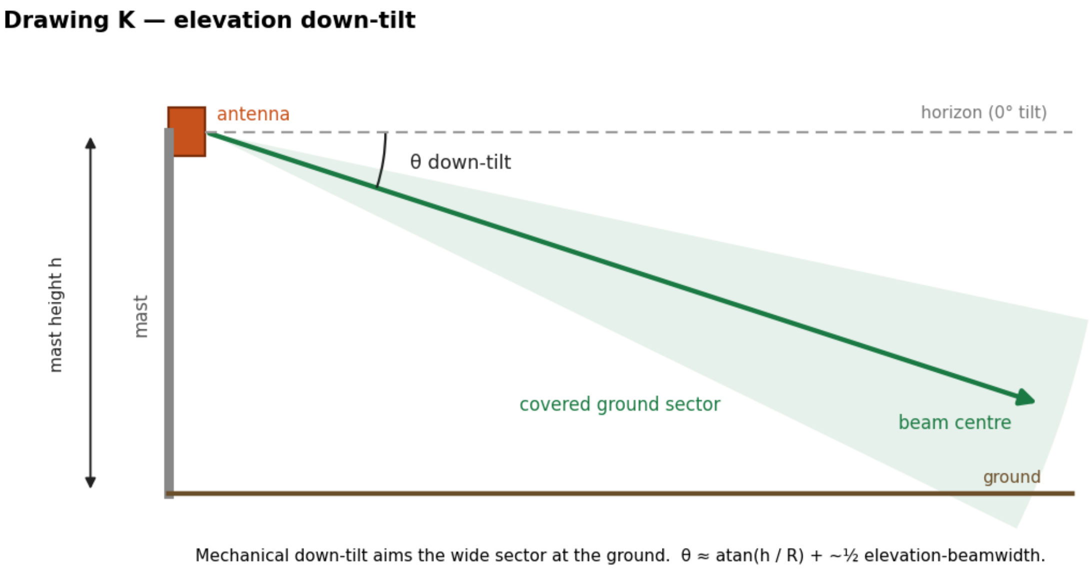
The full 327 mm element is longer than an A4 sheet, so mark it with a tape using these cumulative distances rather than a printed strip (more accurate — printers scale).

Feature	Measure from	Distance
Dipole tube total cut	—	698 mm of Ø6 tube

Feature	Measure from	Distance
Bend 1	one end of tube	327 mm
Bend 2	one end of tube	349 mm
Bend 3	one end of tube	676 mm
Feed gap (centre)	centre of lower long side	12 mm, centred
Reflector size	—	484 × 484 mm
Grid rod pitch	first rod	40 mm each (≤ 69 mm)
Dipole–reflector spacing	reflector face	138 mm (0.20λ)

5 Coverage tuning: down-tilt & stacking

5.1 Elevation down-tilt



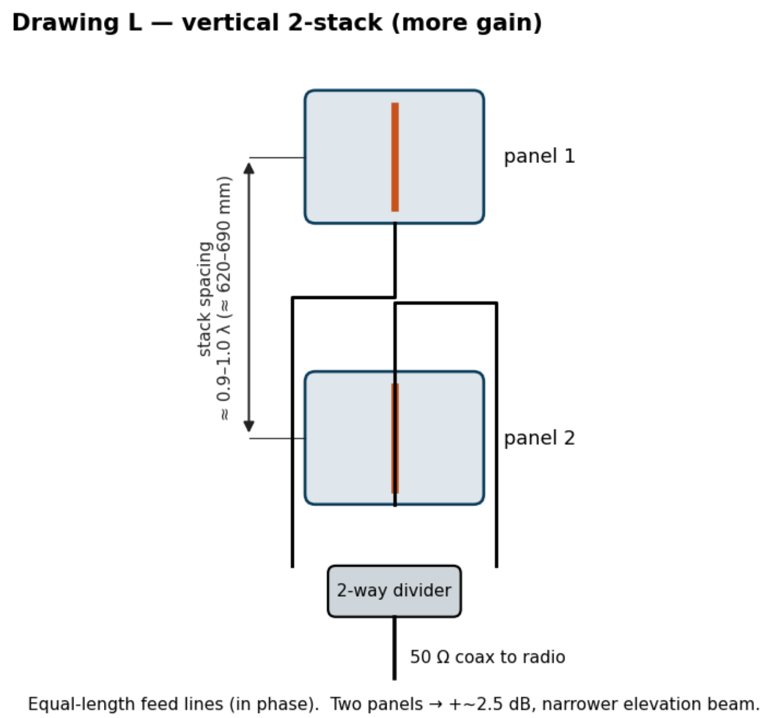
Drawing K — tilting the panel nose-down aims the wide sector onto the ground below the mast.

From a mast of height *h* covering out to range *R*, set the mechanical down-tilt to θ ≈ **atan(h / R) + ~½ of the elevation beamwidth**. Tilt the whole panel on its bracket (shim it nose-down). Typical values:

Mast height	Coverage range R	Down-tilt θ
10 m	100 m	~6–9°
10 m	300 m	~2–5°
20 m	500 m	~2–4°
30 m	1 km	~2–3°

Down-tilt mostly matters close in (high mast, short range); for long, near-horizon links keep θ small. Mechanical tilt is simplest here; commercial panels also offer electrical tilt.

5.2 Stacking two sector panels (more gain)



Drawing L — two of these sector panels stacked vertically and fed in phase.

Stacking **two identical sector panels vertically** keeps the wide ~120° azimuth sector but **narrows the elevation beam**, adding about **+2.5 dB** of gain (so ~7.5 → ~10 dBi) and concentrating energy toward the horizon for longer reach.

- **Spacing:** ~0.9–1.0λ centre-to-centre (≈ 620–690 mm). Wider → more gain but elevation sidelobes grow.
- **Feed:** a **2-way power divider** with **equal-length** coax tails to each panel so they drive in phase; one 50 Ω coax to the radio. A Wilkinson or a λ/4 (75 Ω) matching pair both work.
- **Both panels identical** (same dimensions, same orientation, same polarisation) and co-planar, facing the same way.
- Want even more? A **4-panel vertical stack** adds ~+5–6 dB (~13 dBi) but needs a 4-way harness and a taller frame.
- Stacking does **not** widen the azimuth sector — for a wider arc use two panels splayed in azimuth (each covering an adjacent sector) instead of stacked.

Disclaimer. Dimensions are a computed engineering starting point for a 433.92 MHz reflector-backed dipole. Drawings are to-scale schematics; radiation patterns are **illustrative analytic models** shaped to the target beamwidths; comparison gains/beamwidths are approximate typical values — none are NEC-simulated or anechoic-measured. The 1:1 template only prints true-scale if your printer is set to 100% and the calibration bar measures 100 mm. Verify in NEC and tune on a VNA. Observe local ISM-band power/antenna regulations.